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Inventor(s)

Das Putham Das; Rahul et al.

MICROPAYMENTS TRANSACTION SYSTEM

Abstract

An improved payment system is provided for approving micropayments during interactive services engaged by consumers. The payment system provides a stored list of multiple sources of payment entered by the consumer. A transaction processor receives micropayment transaction payment requests and uses a financial database to approve or deny the request before submitting the payment request to the corresponding cash account or credit account issuer. The improved payment system may be useful in reducing latency due to waiting for payment approvals and may be useful in reducing transaction costs.

Inventors: Das Putham Das; Rahul (Alexandria, VA), Prakash; Mayank (Singapore, SG), Gulati; Nimit (Singapore, SG), Gelli; Sandeep (Singapore, SG)

Applicant: MASTERCARD ASIA/PACIFIC PTE. LTD. (Singapore, SG)

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Background/Summary

BACKGROUND

[0001] The present inventions relate generally to financial transaction processing, and more particularly, to processing micropayment transactions.

[0002] Modern consumer payment systems commonly utilize sophisticated transaction systems involving one bank associated with the consumer, another bank associated with the merchant and a transaction processor that facilitates payments from the consumer bank to the merchant bank. While these systems are widely used, well-known and quite efficient, conventional transaction systems have costs associated with such systems that can be attributed to each individual transaction. In addition, each transaction through such a system takes a certain amount of time to be completed. In most situations, the individual transaction costs and processing time are minimal and do not cause any significant impediments to using such a system for ordinary purchases between consumers and merchants.

[0003] However, in some situations, transaction costs and processing time can become a concern for conventional modern consumer payment systems. One such situation involves micropayments between a consumer and a merchant. Although various monetary values can be used to define a micropayment, payments between a consumer and a merchant involving more than \$10 are usually considered to be conventional payments and are not considered to be micropayments. That is, when the purchase price of a good or a service being sold to a consumer by a merchant is more than \$10, the transaction costs associated with the transaction are minimal relative to the purchase price of the good or service, and therefore, the transaction costs raise few concerns for such purchases. On the other hand, when the purchase price is smaller (i.e., less than at least \$10), the individualized cost of the transaction relative to the purchase price itself can be noticeably higher. This may limit the adoption of conventional modern consumer payment systems for such transactions.

[0004] Additionally, the amount of time that it takes to complete an individual transaction can be an impediment in some cases. Although conventional modern consumer payment systems are relatively fast in completing transactions, in some situations the consumer may have a greater need for transactions to be completed in real-time (i.e., nearly instantaneously). One example of this is computer gaming where the game allows purchases to be made while playing the game and where the purchase affects how the game is played. Not only is the speed of transaction processing critical for computer gaming with in-game purchases, but the value of such purchases is usually in the range of a micropayment. Thus, these types of transactions can be challenging for conventional modern consumer payment systems to complete efficiently and satisfactorily.

SUMMARY

[0005] An improved payment system is described. The improved system may be particularly useful for decreasing the amount of time needed to approve payment requests for micropayments authorized by consumers during interactive services. Transaction costs may also be reduced. The consumer enters multiple sources of payment in a stored list. The sources of payment may include cash accounts or credit accounts. When the consumer authorizes a transaction, a payment request is received by a transaction processor. The transaction processor reviews the payment request by accessing a financial database and approves or denies the payment request before the request is submitted to the consumer's bank for final approval. The financial database may include past financial transactions, past cash account balances or past credit account balances. The invention may also include any other aspect described below in the written description or in the attached drawings and any combinations thereof.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

[0006] The invention may be more fully understood by reading the following description in

conjunction with the drawings, in which:

[0007] FIG. 1A illustrates a game with in-game purchases;

[0008] FIG. 1B illustrates a payment screen with multiple sources of payment;

[0009] FIG. 1C illustrates a purchase history screen;

[0010] FIG. 2 illustrates one embodiment of a micropayment transaction system;

[0011] FIG. 3 illustrates another embodiment of a micropayment transaction system; and

[0012] FIG. 4 illustrates another embodiment of a micropayment transaction system.

DETAILED DESCRIPTION

[0013] Referring now to the figures, and particularly FIG. 1A, an electronic game screen **10** is shown. As shown, the electronic game **10** allows purchases **12** to be made while play continues or while the game **10** is temporarily paused (i.e., during a game session). Typically, such games **10** allow different game items **14** to be purchased at various prices. Typically, in-game purchases **12** (i.e., micropayment transactions) may be used to upgrade electronic tools **14** in the game **10** or for other upgrades or may be used to enable continued playing of the game **10**. In order to facilitate in-game purchases **12**, a payment screen **16** like FIG. 1B may be provided that the consumer **26** may access and set up for completing such in-game purchases **12**. As shown, the payment screen **16** may include a stored list **18** of multiple sources of payment **20, 22** associated with the consumer **26** for paying for in-game purchases **12**. Although numerous types of payment sources **20, 22** may be included in the payment method list **18**, it is preferred for the list **18** to include sources of payment **20, 22** that correspond to both cash accounts **20** and credit account issuers **22**, and more preferably, at least one of the payment sources **20, 22** may be associated with a corresponding credit account issuer **22**. Examples of a cash account **20** include a debit card or a savings account in which the consumer has deposited cash that may be used to pay for purchases using cash payment services like ACH transfers. By contrast, an example of a credit account **22** includes a credit card provided to the consumer **26** by an issuer **32** (also referred to as the consumer's bank **32**). In this case, the issuer **32** approves consumer purchases using credit which the consumer **26** agrees to pay back at a later time. As shown in FIG. 1C, it may also be desirable to provide the consumer **26** with a purchase history screen **24** to show how much the consumer **26** has spent using the list **18** of payment sources **20, 22**. Although it is understood that electronic games **10** like FIG. 1A are one type of interactive service **10** that may allow consumers **26** to make micropayment transactions while the consumer **26** is interactively engaged with the interactive service **10**, there are many other interactive services **10** besides electronic games **10** that may allow micropayment transactions while being engaged with the service.

[0014] In a conventional modern consumer payment system, a consumer **26** initiates a request to pay a merchant **28** for a service or good by providing the merchant **28** with information about an account **20, 22** assigned to the consumer **26**, such as a credit card or debit card number **20, 22**. Typically, the merchant **28** then forwards the payment request to an intermediary like the merchant's bank **34** (e.g., an acquirer **34**) or a payment processor **30** who handles payment requests from numerous merchants **28**. The payment request is then forwarded to the consumer's bank **32** who manages the consumer's account **20, 22** from which the payment will be made. After reviewing the payment request and various details associated therewith (e.g., account status, identity information, amount of payment request, etc.), the consumer's bank **32** approves or denies the payment request and sends the payment decision back to the intermediary **30, 34** for forwarding to the merchant **28**. However, as noted above, conventional payment systems of this type may be inadequate for micropayments, and especially where the micropayment is being made during engagement with an interactive service **10**.

[0015] Thus, turning to FIGS. 2-4, embodiments of an improved system of processing micropayment transactions is provided. In this system, when a consumer **26** initiates a micropayment transaction payment request as described above in connection with FIGS. 1A-1C, the payment request is forwarded by the merchant **28** to a transaction processor **30**. As shown, the

payment request will typically be forwarded to an acquirer **34** who then forwards the request to the transaction processor **30**. As noted above, the payment request requests payment from one of the multiple sources of payment **20, 22** in the stored list **18** of payment sources **20, 22**. The transaction processor **30** then reviews a financial history associated with the consumer **26** which is stored in a financial database and which may include past financial transactions, past cash account balances, past credit account balances and/or other relevant financial information. Unlike conventional payment systems, the transaction processor **30** then approves (or denies) the payment request before submitting the payment request to the consumer's bank **32** (i.e., the cash account **20** or credit account issuer **22**) for approval. One reason that conventional payment systems do not operate in this way is that the transaction processor **30** assumes a financial risk in this scenario because when the transaction processor **30** forwards the payment request to the consumer's bank **32** after the transaction processor **30** has already sent an approval to the merchant **28** it is possible that the consumer's bank **32** will deny the payment request. This means that the transaction processor **30** could be required to make the payment directly itself and accept a financial loss from the transaction.

[0016] Despite this financial risk for the transaction processor **30**, the improved payment system can significantly speed up approvals for payment requests since one of the most time-consuming steps in conventional payment systems is forwarding the payment request to the consumer's bank **32** for approval. This is particularly the case where the source of payment corresponds to a credit account issuer **22, 32**. One reason for this is that credit account issuers **22, 32** provide credit services to consumers **26** and assume certain financial risks themselves of nonpayment by the consumer **26**. Thus, credit account issuers **22, 32** typically require a variety of financial checks before approving payment requests which can significantly increase the time it takes to provide a payment request approval to the merchant **28**.

[0017] The improved payment system may be especially useful where the consumer **26** is engaging an interactive service **10** like an electronic game **10** during a session of the service **10**. In this case, the session is typically stalled while the interactive service **10** waits for the transaction to be completed and depends on the merchant **28** or the acquirer **34** receiving approval of the transaction. Thus, the system is especially useful where the transaction and approval are occurring during a session of an interactive service **10**. The improved service may also be more useful where the transactions are micropayment transactions (e.g., less than \$10 or \$5) since these are more likely to be the types of transactions to occur during a session with an interactive service **10**. By utilizing this payment system for only micropayment transactions, the financial risk assumed by the transaction processor **30** is also minimized.

[0018] The improved payment system also offers the ability to easily switch between different payment sources **20, 22**, potentially by both the consumer **26** and the transaction processor **30**, in order to accommodate various desires. For example, it may be desirable for the consumer **26** to be provided with the ability to sort the multiple sources of payment **20, 22** in order of preference. In this case, the transaction processor **30** selects one of the multiple sources of payment **20, 22** based on the order of preference set by the consumer **26**. However, it is preferred for the transaction processor **30** to be allowed to incorporate other factors as well into the selection process of which payment source **20, 22** to use for a particular transaction. Thus, while the consumer **26** may select the order of preference, the transaction processor **30** may autonomously select which payment source **20, 22** is actually used for approving a transaction, and this may involve the transaction processor **30** selecting a payment source **20, 22** that is lower on the list **18** based on various considerations. For example, it may be desirable for the transaction processor **30** to select payment sources **20, 22** corresponding to a cash account **20** first before selecting a payment source corresponding to a credit account **22** since the transaction processor **30** may assume more financial risk when approving a payment from a credit account **22**. In this scenario, the transaction processor **30** may be able to quickly check the cash balance of cash accounts **20** in the payment list **18** to

determine which cash accounts **20** have a positive balance. This would allow the transaction processor **30** to select one of the cash accounts **20** from the list **18** with a positive balance before selecting a credit account **22** in order to reduce the financial risk for the transaction processor **30**. Preferably, the transaction processor **30** verifies the account balance of a cash account **20** selected for payment before the transaction payment request is approved. In most cases, this will not slow down the approval process significantly because cash balance checks can be done quite quickly compared to other types of account checks. It may also be desirable for the list **18** of payment sources **20**, **22** to include a default payment source that is associated with a corresponding cash account **20** so that payments are made from the default cash account **20** before selecting a different cash account **20** or a credit account **22**. Further, it may be desirable for the transaction processor **30** to only select one of the credit accounts **22** when none of the cash accounts **20** in the list **18** have a positive cash balance.

[0019] The transaction processor **30** may also utilize more sophisticated analysis techniques when selecting which payment source **20**, **22** from the list **18** to use for a particular transaction. For example, because the improved payment system may be particularly useful for interactive services **10**, shorter latency in approving transactions may be especially valuable. Thus, the transaction processor **30** may determine a predicted latency for each of the payment sources **20**, **22** in the stored list **18** and may select a payment source **20**, **22** from the list **18** with the lowest predicted latency. Where micropayments are involved, transaction costs may also be an important factor. Thus, the transaction processor **30** may determine a predicted transaction cost for each of the payment sources **20**, **22** in the stored list **18** and may select a payment source **20**, **22** from the list **18** with the lowest predicted transaction cost. Since the transaction processor **30** may assume a financial risk when approving certain transactions, it may also be desirable for the transaction processor **30** to determine a predicted fraud risk for each of the payment sources **20**, **22** in the list **18** and may select a payment source **20**, **22** from the list **18** with the lowest predicted fraud risk. Such determinations may be made for both the cash accounts **20** and the credit accounts **22** in the list **18** or they may only be performed for certain types of accounts (e.g., credit accounts **22**).

[0020] In order to reduce transaction costs and speed up the approval process, it may also be desirable for the transaction processor **30** to accumulate transactions together for submission to the consumer's bank **32**. For example, when the consumer **26** is engaged with an interactive service **10** and authorizes a micropayment while engaged with the service **10**, it may be expected that the consumer **26** will also authorize additional micropayment transactions within a short period of time while engaged with the service **10**. Thus, in this scenario it may be desirable for the transaction processor **30** to combine a plurality of micropayment transactions together and submit a single transaction payment request to the consumer's bank **32**. This may be particularly useful when the payment source **20**, **22** is a credit account **22** and the consumer's bank **32** is a credit account issuer **32** since these types of accounts usually have higher transaction costs for each transaction. Where the consumer **26** is engaged in a session of an interactive service **10** and the micropayment transactions are being submitted to a credit account issuer **32**, it may be desirable for the transaction processor **30** to combine multiple transactions together and submit a single transaction payment request to the credit account issuer **32** for approval after the session is complete. This has the potential to significantly increase the speed of transaction approvals during an interactive session **10** since the more time-consuming credit account **22** approval process with the consumer's bank **32** only occurs after the session **10** is already complete.

[0021] It may also be desirable for the improved payment system to provide limits both for the consumer **26** and the transaction processor **30**. For example, the consumer **26** may be allowed to set a stored budget limit for micropayment transactions. This may be useful since such transactions may occur during a session of an interactive service **10** and it may be difficult for the consumer **26** to keep track of every transaction authorized by the consumer **26**. Thus, when previously approved micropayment transactions reach the stored budget limit, the transaction processor **30** may deny

additional payment requests. The transaction processor **30** may also set a micropayment transaction limit. In this case, the transaction processor **30** may revert to conventional approval processes when payment requests exceed the micropayment transaction limit. When this happens, the transaction processor **30** may submit the payment request to the consumer's bank **32** (e.g., a credit account issuer **32**) for approval before approving the payment request.

[0022] It is understood that the described improved payment system is intended to operate autonomously on programmed computer systems utilizing computer algorithms such that the system may be implemented by one or more computer processors (e.g., in a server system) executing computer-executable instructions stored on a non-transitory computer-readable storage medium. Thus, for example, in the case of the transaction processor **30** it is unnecessary for human beings to make the required determinations, submissions, approvals, etc. This autonomous design makes the improved payment system scalable to a level that would be impractical if human beings were to attempt to perform the steps required by the system. While it is understood that various human beings may provide inputs to the system and may adjust parameters that control how the system operates, the improved payment system is intended to have the capability of processing many thousands of transactions in short periods of time (e.g., seconds or less) that would be impossible to accomplish with human intervention in each transaction.

[0023] While preferred embodiments of the inventions have been described, it should be understood that the inventions are not so limited, and modifications may be made without departing from the inventions herein. While each embodiment described herein may refer only to certain features and may not specifically refer to every feature described with respect to other embodiments, it should be recognized that the features described herein are interchangeable unless described otherwise, even where no reference is made to a specific feature. It should also be understood that the advantages described above are not necessarily the only advantages of the inventions, and it is not necessarily expected that all of the described advantages will be achieved with every embodiment of the inventions. The scope of the inventions is defined by the appended claims, and all devices and methods that come within the meaning of the claims, either literally or by equivalence, are intended to be embraced therein.

Claims

1. A system of processing micropayment transactions, comprising: an interactive service engaged by a consumer allowing micropayment transactions while the consumer is interactively engaged with the interactive service; a stored list of multiple sources of payment associated with the consumer for paying the micropayment transactions, wherein each of the multiple sources of payment is associated with a corresponding cash account or credit account issuer and at least one of the multiple sources of payment is associated with a corresponding credit account issuer; a financial database comprising a financial history associated with the consumer comprising past financial transactions, past cash account balances or past credit account balances; and a transaction processor receiving a micropayment transaction payment request from the interactive service in response to the consumer authorizing a micropayment transaction during interactive engagement with the interactive service; the micropayment transaction payment request requesting payment from one of the multiple sources of payment; wherein the transaction processor approves the micropayment transaction payment request based on the financial database before submitting the micropayment transaction payment request to the corresponding cash account or credit account issuer for approval.

2. The system of processing micropayment transactions according to claim 1, wherein the transaction processor approves the micropayment transaction payment request based on the financial database before submitting the micropayment transaction payment request to the corresponding credit account issuer for approval.

3. The system of processing micropayment transactions according to claim 1, wherein the consumer engages the interactive service during a session, the micropayment transactions and approval thereof occurring during the session.
4. The system of processing micropayment transactions according to claim 1, wherein the micropayment transaction payment request is for less than \$5.
5. The system of processing micropayment transactions according to claim 1, wherein the interactive service is an electronic game and the micropayment transaction occurs during a session of the electronic game for an upgrade or continued play.
6. The system of processing micropayment transactions according to claim 1, wherein the stored list of multiple sources of payment is sorted by the consumer in order of preference to be used for the micropayment transactions, the transaction processor selecting one of the multiple sources of payment based on the order of preference.
7. The system of processing micropayment transactions according to claim 1, wherein the stored list of multiple sources of payment comprises a default source of payment, the default source of payment being associated with a corresponding cash account.
8. The system of processing micropayment transactions according to claim 1, wherein the transaction processor selects one of the multiple sources of payment from the stored list for approving the micropayment transaction payment request.
9. The system of processing micropayment transactions according to claim 1, further comprising a predicted latency associated with each of the multiple sources of payment, wherein the transaction processor selects one of the multiple sources of payment with a lowest predicted latency from the stored list for approving the micropayment transaction payment request.
10. The system of processing micropayment transactions according to claim 1, further comprising a predicted transaction cost associated with each of the multiple sources of payment, wherein the transaction processor selects one of the multiple sources of payment with a lowest predicted transaction cost from the stored list for approving the micropayment transaction payment request.
11. The system of processing micropayment transactions according to claim 1, further comprising a predicted fraud risk associated with each of the multiple sources of payment, wherein the transaction processor selects one of the multiple sources of payment with a lowest predicted fraud risk from the stored list for approving the micropayment transaction payment request.
12. The system of processing micropayment transactions according to claim 1, wherein the transaction processor selects one of the multiple sources of payment associated with a corresponding cash account having a positive cash balance before selecting one of the multiple sources of payment associated with a corresponding credit account.
13. The system of processing micropayment transactions according to claim 1, wherein the transaction processor selects one of the multiple sources of payment associated with a corresponding credit account when none of the multiple sources of payment is associated with a cash account having a positive cash balance.
14. The system of processing micropayment transactions according to claim 1, wherein the transaction processor verifies an account balance of a cash account corresponding to a selected one of the multiple sources of payment before approving the micropayment transaction payment request.
15. The system of processing micropayment transactions according to claim 1, wherein the consumer authorizes a plurality of micropayment transactions during interactive engagement with the interactive service, the transaction processor combining the plurality of micropayment transactions and submitting a single transaction payment request to the corresponding cash account or credit account issuer for approval.
16. The system of processing micropayment transactions according to claim 15, wherein the single transaction payment request is submitted to a corresponding credit account issuer for approval.
17. The system of processing micropayment transactions according to claim 16, wherein the

consumer engages the interactive service during a session, and the single transaction payment request is submitted to the corresponding credit account issuer for approval after the session is complete.

18. The system of processing micropayment transactions according to claim 1, further comprising a stored budget limit set by the consumer for micropayment transactions, wherein the transaction processor denies the micropayment transaction payment request when previous micropayment transaction payment request approvals reach the stored budget limit.

19. The system of processing micropayment transactions according to claim 1, further comprising a micropayment transaction limit set by the transaction processor, wherein the transaction processor submits the micropayment transaction payment request to the corresponding credit account issuer for approval before approving the micropayment transaction payment request when the micropayment transaction payment request exceeds the micropayment transaction limit.

20. The system of processing micropayment transactions according to claim 1, wherein the interactive service is an electronic game and the micropayment transaction occurs during a session of the electronic game for an upgrade or continued play, the micropayment transaction payment request is for less than \$5, the transaction processor selects one of the multiple sources of payment from the stored list for approving the micropayment transaction payment request, further comprising a predicted latency associated with each of the multiple sources of payment, wherein the transaction processor selects one of the multiple sources of payment with a lowest predicted latency from the stored list for approving the micropayment transaction payment request, and the transaction processor selects one of the multiple sources of payment associated with a corresponding cash account having a positive cash balance before selecting one of the multiple sources of payment associated with a corresponding credit account.
